

Image Restoration with Universal and Routed Autoencoders and Style-Conditioned Face-to-Sketch Generation

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Abstract—This project implements four connected generative imaging workflows: a universal denoising autoencoder, a hard-routed system of specialist autoencoders, a jointly trained soft mixture, and a style-conditioned paired conditional GAN. The first three share the Oxford-IIIT Pet split and corruption definitions; the fourth uses paired FS2K photographs and sketches. The implementation uses validation-only selection of hyperparameters and checkpoints, with final model hashes frozen before test evaluation (Section III-B) and deploys verified ONNX inference through a React and FastAPI application started with Docker Compose. The following sections report the recorded final test measurements. On 36,690 deterministic test inputs the corruption classifier reaches 96.8% accuracy (macro F1 0.950). Predicted routing changes mean PSNR by +9.6 dB on salt-and-pepper noise and +8.2 dB on occlusion, but by -1.7 dB on blur, where the soft mixture reaches -0.4 dB. The sketch generator reaches a mean SSIM of 0.45 and produces smooth sketches without stroke detail.

Index Terms—denoising autoencoder, mixture of experts, conditional GAN, ONNX

I. Introduction

Image restoration reconstructs a clean target when the observed image is affected by impulse noise, blur, or missing regions. A universal model offers a single reconstruction path, while specialist models can allocate capacity to individual corruption types. Hard routing introduces classifier mistakes; soft routing permits uncertain inputs to combine several outputs. Paired sketch generation additionally requires a style condition and an adversarial objective. The research questions are whether specialization improves reconstruction, how routing errors change quality, whether soft weights remain balanced, and whether a compact conditional generator responds to the three FS2K styles. These questions require measured results rather than assumed rankings. The contributions are: (i) a runtime-corruption pipeline with deterministic validation and test manifests; (ii) a compressed universal autoencoder, a classifier-routed set of specialists, and a jointly trained soft mixture that share one data protocol; (iii) a style-conditioned paired GAN; (iv) Optuna and MLflow records for every task; and (v) one Docker Compose application exposing all four workspaces through verified ONNX models.

II. Related Research and Design Alternatives

Autoencoders learn compact representations by forcing data through a bottleneck [11], and denoising training maps corrupted inputs to clean targets [1]. Supervised convolutional restoration networks such as DnCNN [12] and encoder-decoder networks with long skip connections such as U-Net [13] are strong baselines. Skip connections were deliberately not used in the restoration autoencoders: they would let the decoder copy uncompressed input features, which the assignment’s bottleneck requirement excludes. The consequence, examined in Section X, is limited texture recovery.

SSIM evaluates local structural agreement [2]; L1 penalizes pixel differences. Combining them provides distinct reconstruction signals, but their weight is selected through validation. The soft system follows the mixture-of-experts idea of Jacobs et al. [14]. Expert imbalance is a known concern [3], [15]; this project uses a dense output mixture with the assignment’s batch-mean balance penalty, not the sparse routing of those works. Conditional GANs [16], [17] and paired translation with a U-Net generator and local (PatchGAN) discriminator [4] motivate Task 4, using FS2K [18]. The compact discriminator here has a 38-pixel receptive field; it is a local discriminator variant rather than the original 70-pixel design.

The pilot objectives in the table are the validation measurements recorded in Section X and the project’s research notes; they are exploratory sequential changes rather than controlled single-factor ablations.

III. Data Preparation and Reproducibility

All images are RGB and resized to 128×128 with bicubic interpolation. The Oxford official trainval membership [5] is split with NumPy random seed 42, yielding 2,944 training and 736 validation images. The official test set contains 3,669 images. The same membership is used for Tasks 1–3; breed labels do not supervise restoration. Source archives and preparation code are retained with manifest hashes. Test images are downloaded and resized during preparation, but never loaded by training or hyperparameter selection.

The official FS2K annotations define 1,058 development pairs and 1,046 test pairs [6]. A stratified 15% validation

TABLE I
Alternatives investigated and the decision taken

Alternative	Evidence or reasoning	Decision
Restoration U-Net with long skips	Bypasses the required bottleneck [13]	Rejected
Dense latent vector (128 values)	Collapsed, nearly constant outputs in validation grids (objective 0.322)	Replaced by spatial latent
8×8 spatial latent (512 values)	Objective 0.281; coarse shapes, large colour error	Replaced by 16×16
GroupNorm in the autoencoder	Per-input statistics; BatchNorm [21] with tracked statistics tested next	BatchNorm adopted
Learning rate 10^{-3}	Saturated outputs in a pilot	Rejected; $\leq 5 \times 10^{-4}$ searched
Bilinear resize + conv	Compared with PixelShuffle decoder [9]	PixelShuffle + ICNR [10]
Sparse top- k gating [3]	Only three experts; dense blend is differentiable and matches the brief	Dense softmax gate
Per-batch entropy regulariser [15]	Not evaluated; assignment’s squared deviation kept	Squared deviation from 1/4

split with seed 42 yields 899 training and 159 validation pairs. Training style counts are 303/298/298 and validation counts are 54/52/53. Style labels are read from annotations, not inferred from folder names. The official photograph-to-sketch filename mapping is preserved. Horizontal flips are applied identically to both images in a training pair.

A. Runtime Corruptions

Every clean training image receives a freshly sampled condition. The universal model samples each of four conditions with equal probability; classifier and gate batches enforce equal counts. Salt-and-pepper probability is uniform in $[0.02, 0.15]$, with selected RGB pixels set to black or white equiprobably. Blur uses a finite separable Gaussian kernel of size 3, 5, or 7 and sigma uniform in $[0.5, 2.5]$. Occlusion samples one to three disjoint black rectangles whose joint area approximates a sampled fraction in $[0.10, 0.35]$. Disjoint horizontal bands prevent overlap; random widths and positions vary mask geometry. This restriction should be considered when interpreting generalization to arbitrary real occlusions.

Validation and test manifests persist condition, seed, blur settings, and mask coordinates. Each image has one clean condition and nine corrupted conditions. Pixel

TABLE II
Fixed validation and test severity definitions

Condition	Low	Medium	High
Salt probability	.03	.08	.15
Blur (k, σ)	(3, .7)	(5, 1.5)	(7, 2.5)
Occlusion area	10%	20%	35%
Rectangle count	1	2	3

rounding can slightly change the realized mask area, which is recoverable from stored coordinates. Training corruptions are not saved as image copies.

B. Evaluation Protocol and Test-Set Use

All hyperparameters, architectures and checkpoints were selected using the validation split only. The model files evaluated in this report were frozen by SHA-256 hash before the final test run; the hashes and selected epochs are stored in the results summary. The official test split was also evaluated for two earlier model generations during development, so it is a held-out reporting set rather than a set seen for the first time. Architecture, search and checkpoint decisions used validation measurements only.

IV. Task 1: Universal Restoration

Three stride-two encoder convolutions have widths $b, 2b, 4b$, reducing the image from 128 to 16 pixels. LeakyReLU and local residual blocks refine features, with BatchNorm after stage convolutions, using tracked running statistics during inference. A 1×1 convolution compresses to $d/256$ channels on a 16×16 grid. A corresponding expansion and three learned convolution and PixelShuffle upsampling stages [9], initialized with phase-matched ICNR kernels [10] reconstruct RGB with sigmoid output. The latent search contains $d \in \{2048, 4096, 8192\}$ values, all smaller than the 49,152-value input. Residual blocks refine features within a stage; they never bypass the compressed representation. No skip connection copies uncompressed features to the decoder.

$$\hat{x} = D(E(\tilde{x})), \quad \mathcal{L}_{rec} = \alpha \|\hat{x} - x\|_1 + (1 - \alpha)(1 - \text{SSIM}(\hat{x}, x)). \quad (1)$$

SSIM uses an 11×11 Gaussian window with sigma 1.5, valid convolution, $C_1 = 0.01^2$, and $C_2 = 0.03^2$. It is averaged over RGB channels and windows. The validation objective fixes $\alpha = 0.8$ for comparability across trials; the training weight is tuned.

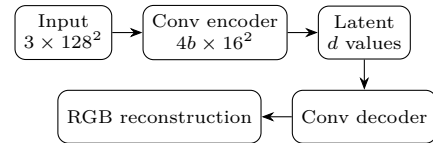


Fig. 1. Task 1 compressed restoration path. Tasks 2 and 3 reuse this expert architecture.

V. Task 2: Classification and Hard Routing

A four-stage convolutional classifier uses global average pooling, dropout, and a four-logit linear head. Multiclass cross entropy supervises the corruption label generated by the data loader. Three independently optimized autoencoders learn salt, blur, and occlusion restoration. A shared architecture search scores the average validation objective across all three specialists.

$$r = \arg \max_k C(\tilde{x})_k, \quad \hat{x} = \begin{cases} \tilde{x}, & r = 0, \\ A_r(\tilde{x}), & r \neq 0. \end{cases} \quad (2)$$

Oracle routing selects the branch using the true manifest label; predicted routing uses classifier argmax. Comparing these modes isolates the contribution of routing mistakes. Clean oracle inputs have an exact identity bypass; a misclassified clean input can be unnecessarily altered in predicted mode. Accuracy, macro precision/recall/F1, per-class metrics, a normalized confusion matrix, and individual routing failures are included after evaluation.

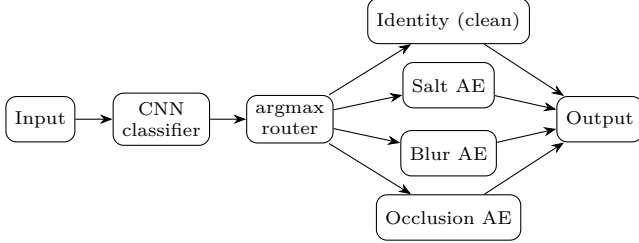


Fig. 2. Task 2 hard routing: the classifier selects exactly one branch; clean inputs bypass restoration.

VI. Task 3: Joint Soft Mixture

The gate is initialized from the trained classifier and the three experts from Task 2. An initial two-epoch warm-up trains only the gate with experts frozen; the experts are then unfrozen and the complete system is fine-tuned jointly. Both stages use one Adam learning rate, searched in $[10^{-5}, 3 \times 10^{-4}]$ (selected 2.7×10^{-4} , below the classifier’s 7.9×10^{-4} and slightly below the specialists’ 3×10^{-4}); it is not reduced a second time at unfreezing.

$$w = \text{softmax}(G(\tilde{x})/\tau), \quad (3)$$

$$\hat{x} = w_0 \tilde{x} + \sum_{k=1}^3 w_k A_k(\tilde{x}), \quad (4)$$

$$\mathcal{L}_{\text{joint}} = \alpha \mathcal{L}_1 + (1 - \alpha)(1 - \text{SSIM}) + \lambda_{\text{cls}} \mathcal{L}_{\text{CE}} + \lambda_{\text{bal}} \sum_{k=0}^3 (\bar{w}_k - 1/4)^2. \quad (5)$$

The balance target assumes balanced training batches. A temperature above one softens weights; a lower value sharpens them. Mean weights per condition and severity, distributed and dominant examples, and branches receiving under 1% mean weight are reported. The inference graph evaluates all branches, so it does not inherit sparse-compute speed advantages.

VII. Task 4: Style-Conditioned Face-to-Sketch GAN

A five-level U-Net downsamples from 128 to 4 pixels and upsamples through skip-connected decoding stages. A learned style embedding is spatially broadcast and concatenated with the photograph. The discriminator separately embeds the same style, concatenating it with photograph and sketch. Its three stride-two convolutional stages followed by a stride-one logit convolution produce local real/fake predictions. The refined generator uses

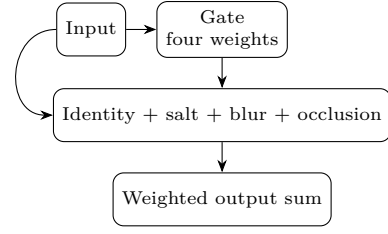


Fig. 3. Soft routing; hard routing instead uses argmax and executes one selected branch.

BatchNorm with tracked inference statistics in encoder and intermediate decoder stages, learned convolution and PixelShuffle upsampling, and phase-matched ICNR initialization [9], [10]. The discriminator normalizes its second and third downsampling stages. Dropout is applied to early decoder stages.

$$\mathcal{L}_D = \frac{1}{2} [\text{BCE}(D(x, y, s), 1) + \text{BCE}(D(x, G(x, s), s), 0)], \quad (6)$$

$$\mathcal{L}_G = \text{BCE}(D(x, G(x, s), s), 1) + \lambda_1 \|G(x, s) - y\|_1. \quad (7)$$

Discriminator real/fake losses and generator adversarial/reconstruction losses are logged separately. The same fixed validation photographs are saved every five epochs. Only the generator is deployed. Cross-style outputs on one photo are qualitative evidence; paired metrics apply to its annotated style only.

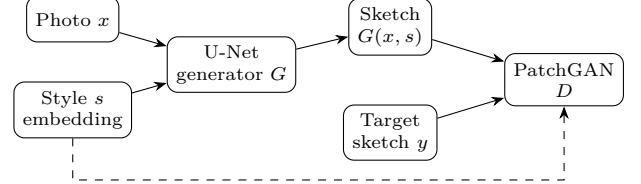


Fig. 4. Task 4 conditional GAN. The style embedding enters both G and D (dashed); D outputs a map of local real/fake scores.

VIII. Hyperparameter Search and Training

Optuna TPE samples configurations and median pruning stops unpromising individual-model trials [7], [20]. Short trials use 3 epochs, 40 training batches and 40 validation batches for the new restoration and GAN studies; the historical classifier study retains its original budget; each study records its actual budget and trial outcomes. The shared specialist study evaluates all three specialists and does not use intermediate pruning. Selected restoration configurations are trained from scratch on the full training split; the classifier is retained and the refined GAN is independently tuned and trained from scratch. Adam [19] is used; GAN optimizers use $(\beta_1, \beta_2) = (0.5, 0.999)$. CUDA runs use mixed precision; gradient norms are clipped to five. Checkpoints include optimizer, scaler, random state, epoch, configuration, and data hashes for recovery.

TABLE III
Search spaces (selected model generation)

Study	Searched parameters
Universal	LR 10^{-4} -.0005; batch 16/32/64; base 16/24/32; latent 2048/4096/8192; dropout 0-.08; α .6-.95
Classifier	LR 10^{-4} -.003; batch 16/32/64; base 8/16/24; dropout 0-.3; weight decay 10^{-6} - 10^{-3}
Specialists	Shared LR, batch, base, latent, dropout, α as universal
Soft	LR 10^{-5} -.0003; batch 16/32/64; τ .5-2; λ_{cls} .02-.3; λ_{bat} .001-.1; α .6-.95
GAN	Generator LR 10^{-4} -.0005; discriminator LR .0001-.001; batch 16/32/64; base 16/24/32; dropout 0-.08; embedding 4/8/16; λ_1 50-150

MLflow records configurations, losses, validation metrics and sample artifacts [8]. Exploratory pilots used a free Colab T4 without mounting Google Drive. After its free GPU quota was reached and temporary new weights were lost, improvement searches and final training ran on the laptop CPU with persistent local checkpoints, optimizer/RNG state and SQLite databases. Earlier downloaded models and records were preserved separately. Limited studies cannot support a claim of globally optimal architectures.

IX. Evaluation and Results

PSNR is computed per image with range one and then averaged. SSIM and L1 use the same preprocessing for every system. Clean identity PSNR is capped at 100 dB through the numeric floor on mean-squared error. Each pet test image has ten deterministic input conditions (36,690 inputs in a full evaluation). Per-image CSVs preserve routing labels, errors and scores. Subset evaluations are explicitly labelled and must not be represented as full test results.

Status: Measured test results; complete evaluation.

A. Task 1: Universal model

TABLE IV
Task 1: Universal model

Condition	Severity	n	L1	PSNR	SSIM
clean	clean	3669	0.0338	26.61	0.8216
salt	low	3669	0.0371	25.76	0.7695
salt	medium	3669	0.0416	24.75	0.7206
salt	high	3669	0.0478	23.58	0.6656
blur	low	3669	0.0333	26.69	0.8212
blur	medium	3669	0.0359	25.88	0.7827
blur	high	3669	0.0436	24.11	0.6881
occlusion	low	3669	0.0427	23.62	0.7602
occlusion	medium	3669	0.0513	21.85	0.7042
occlusion	high	3669	0.0659	19.80	0.6173

Mean per-input PSNR is 24.26 dB; the corrupted-input baseline is 27.31 dB. Clean identity inputs influence these averages, so condition-specific rows should guide interpretation.

Salt: PSNR changes relative to leaving the corrupted input unchanged are +5.62, +8.88, +10.44 dB

at low/medium/high severity; the fixed reconstruction objective improves in 3 of 3 severity groups.

Blur: PSNR changes relative to leaving the corrupted input unchanged are -5.65, -0.89, -0.24 dB at low/medium/high severity; the fixed reconstruction objective improves in 0 of 3 severity groups.

Occlusion: PSNR changes relative to leaving the corrupted input unchanged are +7.01, +8.62, +9.12 dB at low/medium/high severity; the fixed reconstruction objective improves in 2 of 3 severity groups.

On clean inputs, L1 is 0.03376, SSIM is 0.82163, and the fixed objective is 0.06268. The identity reference has no reconstruction error; any clean-image degradation is therefore a limitation.

universal: examples 1-4



Fig. 5. Task 1: Universal model: representative examples 1-4. Columns: clean target, input, output, absolute error.

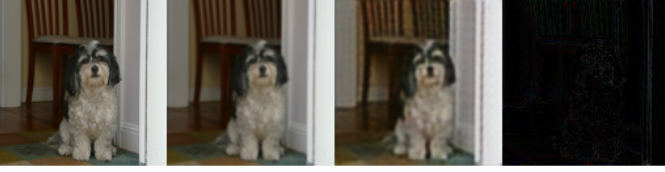
Failure 1: Egyptian_Mau_60, occlusion at high severity, PSNR 8.72 dB and SSIM 0.1795. Large masks hide most of the cat/background; smooth guessed colors fail to recover the scene.

Failure 2: Ragdoll_78, occlusion at high severity, PSNR 8.80 dB and SSIM 0.4455. The mask crosses the face and body; fur detail is lost and the filled region remains blurry.

Failure 3: chihuahua_84, occlusion at medium severity, PSNR 10.02 dB and SSIM 0.4394. Background occlusion

universal: examples 5-8

Clean target Input Output Absolute error
havanese_197 | blur low | PSNR 28.93



shiba_inu_73 | blur medium | PSNR 22.75



basset_hound_24 | blur high | PSNR 24.75

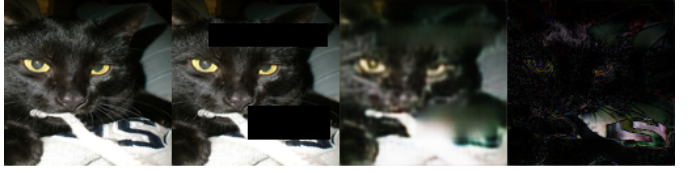


pug_79 | occlusion low | PSNR 21.51

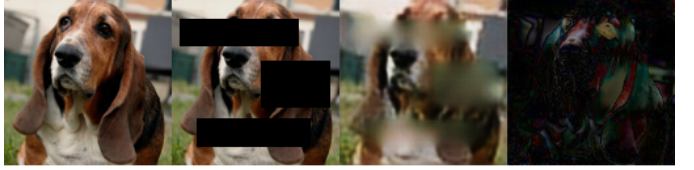


universal: examples 9-12

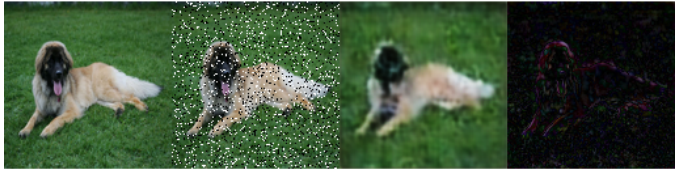
Clean target Input Output Absolute error
Bombay_54 | occlusion medium | PSNR 19.05



basset_hound_53 | occlusion high | PSNR 20.87



leonberger_56 | salt high | PSNR 24.36



yorkshire_terrier_1 | occlusion high | PSNR 21.96



Fig. 6. Task 1: Universal model: representative examples 5-8. Columns: clean target, input, output, absolute error.

Fig. 7. Task 1: Universal model: representative examples 9-12. Columns: clean target, input, output, absolute error.

is filled by a broad smooth band instead of the original textured scene.

Failure 4: shiba_inu_95, occlusion at high severity, PSNR 10.83 dB and SSIM 0.4572. Masks across the head and muzzle produce inaccurate face colors and blurred contours.

B. Task 2: Oracle routing

TABLE V
Task 2: Oracle routing

Condition	Severity	n	L1	PSNR	SSIM
clean	clean	3669	0.0000	100.00	1.0000
salt	low	3669	0.0343	26.40	0.8163
salt	medium	3669	0.0354	26.10	0.8012
salt	high	3669	0.0381	25.46	0.7720
blur	low	3669	0.0338	26.65	0.8299
blur	medium	3669	0.0334	26.63	0.8198
blur	high	3669	0.0405	24.78	0.7234
occlusion	low	3669	0.0438	23.56	0.7654
occlusion	medium	3669	0.0509	21.92	0.7182
occlusion	high	3669	0.0638	19.81	0.6402

Mean per-input PSNR is 32.13 dB; the corrupted-input baseline is 27.31 dB. Clean identity inputs influence

these averages, so condition-specific rows should guide interpretation.

Salt: PSNR changes relative to leaving the corrupted input unchanged are +6.26, +10.23, +12.32 dB at low/medium/high severity; the fixed reconstruction objective improves in 3 of 3 severity groups.

Blur: PSNR changes relative to leaving the corrupted input unchanged are -5.69, -0.15, +0.42 dB at low/medium/high severity; the fixed reconstruction objective improves in 2 of 3 severity groups.

Occlusion: PSNR changes relative to leaving the corrupted input unchanged are +6.95, +8.69, +9.13 dB at low/medium/high severity; the fixed reconstruction objective improves in 2 of 3 severity groups.

On clean inputs, L1 is 0.00000, SSIM is 1.00000, and the fixed objective is 0.00000. The identity reference has no reconstruction error; any clean-image degradation is therefore a limitation.

C. Task 2: Predicted routing

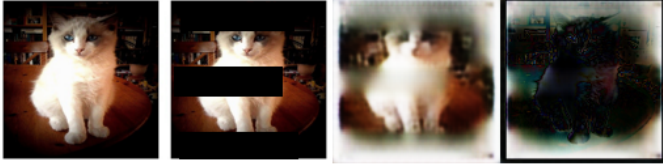
Mean per-input PSNR is 30.91 dB; the corrupted-input baseline is 27.31 dB. Clean identity inputs influence

universal: failures (test)

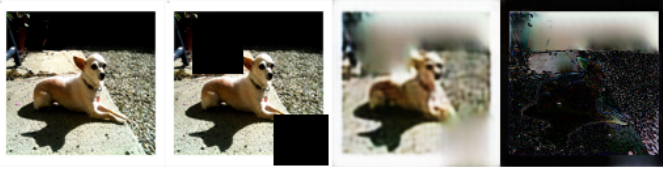
Clean target Input Output Absolute error
Egyptian_Mau_60 | occlusion high | PSNR 8.72



Ragdoll_78 | occlusion high | PSNR 8.80



chihuahua_84 | occlusion medium | PSNR 10.02



shiba_inu_95 | occlusion high | PSNR 10.83



Fig. 8. Task 1: Universal model: four lowest-PSNR examples with distinct source images.

TABLE VI
Task 2: Predicted routing

Condition	Severity	n	L1	PSNR	SSIM
clean	clean	3669	0.0055	87.60	0.9714
salt	low	3669	0.0355	26.27	0.8129
salt	medium	3669	0.0354	26.10	0.8012
salt	high	3669	0.0381	25.46	0.7720
blur	low	3669	0.0318	27.08	0.8415
blur	medium	3669	0.0335	26.61	0.8188
blur	high	3669	0.0406	24.76	0.7227
occlusion	low	3669	0.0438	23.51	0.7662
occlusion	medium	3669	0.0509	21.92	0.7182
occlusion	high	3669	0.0638	19.81	0.6402

these averages, so condition-specific rows should guide interpretation.

Salt: PSNR changes relative to leaving the corrupted input unchanged are +6.14, +10.23, +12.32 dB at low/medium/high severity; the fixed reconstruction objective improves in 3 of 3 severity groups.

Blur: PSNR changes relative to leaving the corrupted input unchanged are -5.25, -0.17, +0.41 dB at low/medium/high severity; the fixed reconstruction objective improves in 1 of 3 severity groups.

Occlusion: PSNR changes relative to leaving the corrupted input unchanged are +6.90, +8.69, +9.13 dB at low/medium/high severity; the fixed reconstruction objective improves in 2 of 3 severity groups.

On clean inputs, L1 is 0.00552, SSIM is 0.97136, and the fixed objective is 0.01014. The identity reference has no reconstruction error; any clean-image degradation is therefore a limitation.

predicted_routing: examples 1-4

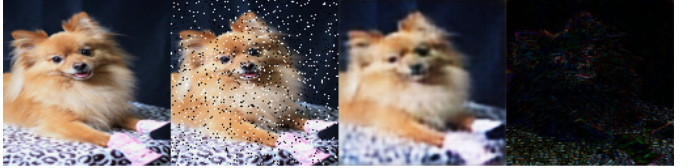
Clean target Input Output Absolute error
basset_hound_36 | clean clean | PSNR 100.00



saint_bernard_62 | salt low | PSNR 26.49



pomeranian_28 | salt medium | PSNR 23.17



havanese_34 | salt high | PSNR 24.15

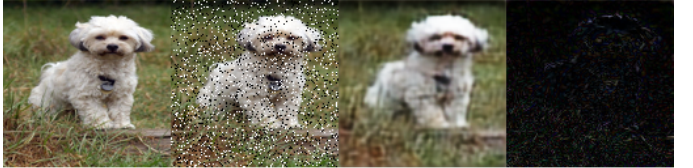


Fig. 9. Task 2: Predicted routing: representative examples 1-4. Columns: clean target, input, output, absolute error.

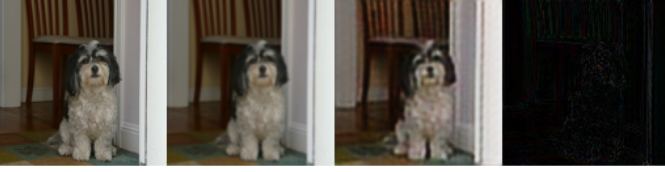
Failure 1: Egyptian_Mau_60, salt at low severity, PSNR 7.10 dB and SSIM 0.2550. The classifier selected occlusion, differing from the true salt condition. Salt noise is misclassified as occlusion on a dark background; the wrong expert produces severe smoothing and false background colors.

Failure 2: Sphynx_23, salt at low severity, PSNR 8.28 dB and SSIM 0.3397. The classifier selected occlusion, differing from the true salt condition. Salt noise is misclassified as occlusion on a dark background; the wrong expert produces severe smoothing and false background colors.

Failure 3: Egyptian_Mau_97, salt at low severity, PSNR 8.65 dB and SSIM 0.3181. The classifier selected occlusion, differing from the true salt condition. Salt

predicted_routing: examples 5-8

Clean target Input Output Absolute error
havanese_197 | blur low | PSNR 28.67



shiba_inu_73 | blur medium | PSNR 23.54



basset_hound_24 | blur high | PSNR 25.45

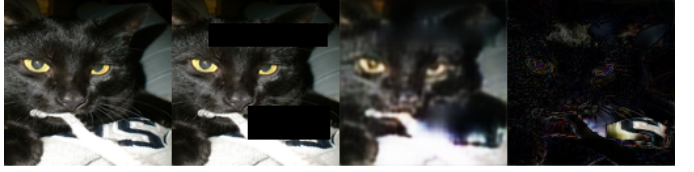


pug_79 | occlusion low | PSNR 22.35

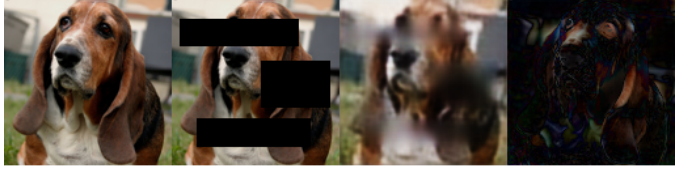


predicted_routing: examples 9-12

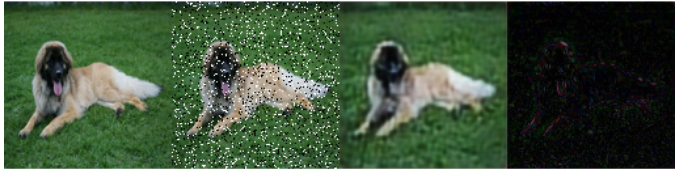
Clean target Input Output Absolute error
Bombay_54 | occlusion medium | PSNR 18.53



basset_hound_53 | occlusion high | PSNR 21.08



leonberger_56 | salt high | PSNR 26.10



yorkshire_terrier_1 | occlusion high | PSNR 22.38



Fig. 10. Task 2: Predicted routing: representative examples 5-8. Columns: clean target, input, output, absolute error.

Fig. 11. Task 2: Predicted routing: representative examples 9-12. Columns: clean target, input, output, absolute error.

noise is misclassified as occlusion on a dark background; the wrong expert produces severe smoothing and false background colors.

Failure 4: Egyptian_Mau_63, salt at low severity, PSNR 8.81 dB and SSIM 0.3762. The classifier selected occlusion, differing from the true salt condition. Salt noise is misclassified as occlusion on a dark background; the wrong expert produces severe smoothing and false background colors.

D. Task 3: Soft mixture

TABLE VII
Task 3: Soft mixture

Condition	Severity	n	L1	PSNR	SSIM
clean	clean	3669	0.0033	50.65	0.9948
salt	low	3669	0.0343	26.50	0.8294
salt	medium	3669	0.0359	26.05	0.8120
salt	high	3669	0.0375	25.64	0.7887
blur	low	3669	0.0225	29.86	0.9091
blur	medium	3669	0.0317	27.16	0.8395
blur	high	3669	0.0388	25.13	0.7344
occlusion	low	3669	0.0402	20.32	0.8600
occlusion	medium	3669	0.0506	21.61	0.7445
occlusion	high	3669	0.0667	19.32	0.6418

Mean per-input PSNR is 27.23 dB; the corrupted-input baseline is 27.31 dB. Clean identity inputs influence these averages, so condition-specific rows should guide interpretation.

Salt: PSNR changes relative to leaving the corrupted input unchanged are +6.37, +10.18, +12.50 dB at low/medium/high severity; the fixed reconstruction objective improves in 3 of 3 severity groups.

Blur: PSNR changes relative to leaving the corrupted input unchanged are -2.48, +0.38, +0.78 dB at low/medium/high severity; the fixed reconstruction objective improves in 2 of 3 severity groups.

Occlusion: PSNR changes relative to leaving the corrupted input unchanged are +3.72, +8.39, +8.64 dB at low/medium/high severity; the fixed reconstruction objective improves in 3 of 3 severity groups.

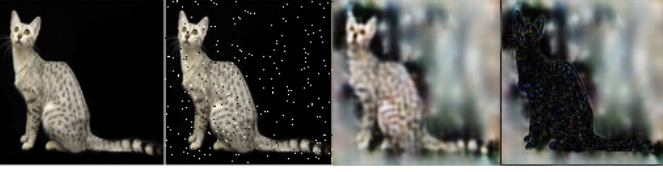
On clean inputs, L1 is 0.00330, SSIM is 0.99477, and the fixed objective is 0.00369. The identity reference has no reconstruction error; any clean-image degradation is therefore a limitation.

Failure 1: Egyptian_Mau_60, occlusion at medium severity, PSNR 11.63 dB and SSIM 0.4195. The mixture

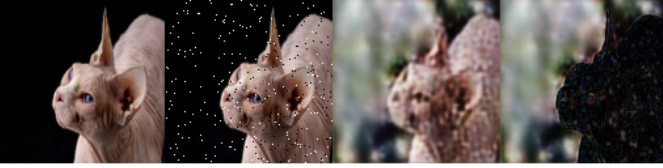
predicted_routing: failures (test)

Clean target Input Output Absolute error

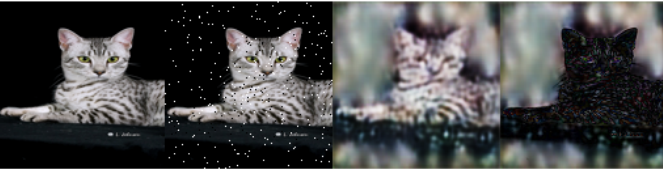
Egyptian_Mau_60 | salt low | PSNR 7.10



Sphynx_23 | salt low | PSNR 8.28



Egyptian_Mau_97 | salt low | PSNR 8.65



Egyptian_Mau_63 | salt low | PSNR 8.81

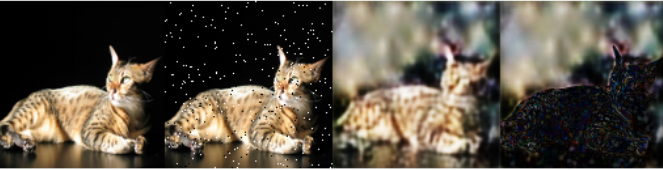


Fig. 12. Task 2: Predicted routing: four lowest-PSNR examples with distinct source images.

fills a masked dark background with incorrect smooth colors.

Failure 2: german_shorthaired_96, occlusion at low severity, PSNR 11.88 dB and SSIM 0.8767. A visible speckled face is partly retained rather than cleaned; the mixture still loses some detail.

Failure 3: staffordshire_bull_terrier_92, occlusion at high severity, PSNR 12.02 dB and SSIM 0.7187. Large masks across the face/body produce a blurred, inaccurate animal reconstruction.

Failure 4: boxer_91, occlusion at high severity, PSNR 12.32 dB and SSIM 0.4844. The large foreground mask destroys body/chair detail; the mixture substitutes smooth approximate colors.

E. Corruption classifier

Overall accuracy is 0.9685. Macro precision, recall and F1 are 0.9552, 0.9452 and 0.9499.

Egyptian_Mau_60: true salt (low), predicted occlusion; predicted-routing PSNR 7.10 dB versus oracle 19.99 dB (difference -12.89 dB). Sphynx_23: true salt (low), predicted occlusion; predicted-routing PSNR 8.28 dB versus oracle 28.56 dB (difference -20.28 dB).

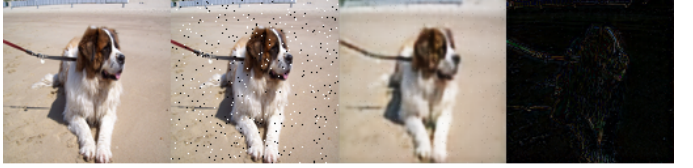
soft: examples 1-4

Clean target Input Output Absolute error

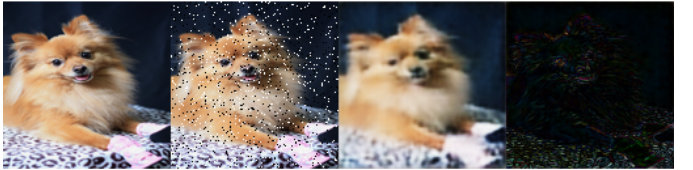
basset_hound_36 | clean clean | PSNR 55.74



saint_bernard_62 | salt low | PSNR 27.04



pomeranian_28 | salt medium | PSNR 23.06



havanese_34 | salt high | PSNR 24.38

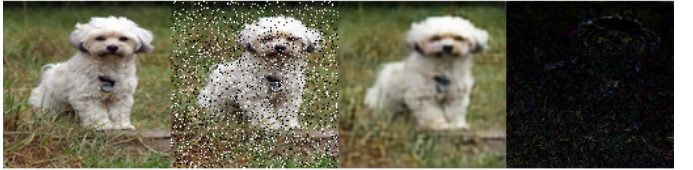


Fig. 13. Task 3: Soft mixture: representative examples 1-4. Columns: clean target, input, output, absolute error.

TABLE VIII
Per-class classifier results

Class	Precision	Recall	F1	n
clean	0.8923	0.8288	0.8594	3669
salt	0.9918	0.9973	0.9945	11007
blur	0.9576	0.9590	0.9583	11007
occlusion	0.9793	0.9956	0.9874	11007

Egyptian_Mau_97: true salt (low), predicted occlusion; predicted-routing PSNR 8.65 dB versus oracle 25.07 dB (difference -16.43 dB). Egyptian_Mau_63: true salt (low), predicted occlusion; predicted-routing PSNR 8.81 dB versus oracle 25.32 dB (difference -16.51 dB). A wrong label changes the selected branch, but its numerical effect can be positive or negative because the specialists are imperfect; classification correctness alone does not establish restoration quality.

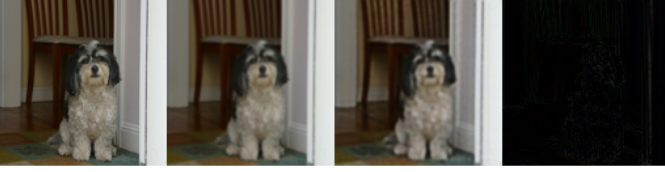
F. Expert weight analysis

Branches below 1% global mean contribution: none.

Examples with dominant weights: Bombay_76 (clean=0.000, salt=1.000, blur=0.000, occlusion=0.000); Egyptian_Mau_60 (clean=0.000, salt=1.000, blur=0.000,

soft: examples 5-8

Clean target Input Output Absolute error
havanese_197 | blur low | PSNR 32.18



shiba_inu_73 | blur medium | PSNR 24.27



basset_hound_24 | blur high | PSNR 25.98

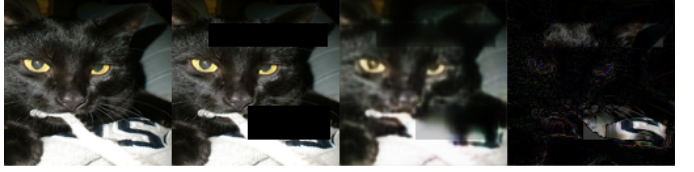


pug_79 | occlusion low | PSNR 16.01

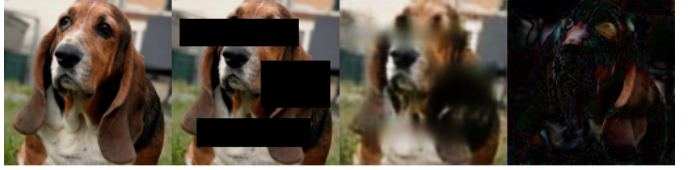


soft: examples 9-12

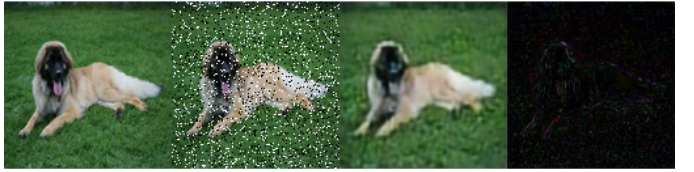
Clean target Input Output Absolute error
Bombay_54 | occlusion medium | PSNR 18.51



basset_hound_53 | occlusion high | PSNR 20.84



leonberger_56 | salt high | PSNR 26.25



yorkshire_terrier_1 | occlusion high | PSNR 22.23



Fig. 14. Task 3: Soft mixture: representative examples 5-8. Columns: clean target, input, output, absolute error.

Fig. 15. Task 3: Soft mixture: representative examples 9-12. Columns: clean target, input, output, absolute error.

TABLE IX
Mean soft branch weights

Class	Severity	Clean	Salt	Blur	Mask
clean	clean	0.911	0.016	0.029	0.043
salt	low	0.044	0.953	0.000	0.003
salt	medium	0.000	1.000	0.000	0.000
salt	high	0.000	1.000	0.000	0.000
blur	low	0.558	0.000	0.407	0.035
blur	medium	0.291	0.000	0.690	0.019
blur	high	0.294	0.000	0.687	0.019
occlusion	low	0.594	0.009	0.006	0.391
occlusion	medium	0.112	0.001	0.001	0.886
occlusion	high	0.004	0.000	0.000	0.996

occlusion=0.000); these examples demonstrate gate concentration, not automatically improved restoration.

Examples with distributed weights: shiba_inu_9 (clean=0.322, salt=0.000, blur=0.341, occlusion=0.337); pug_74 (clean=0.357, salt=0.349, blur=0.000, occlusion=0.294); these examples demonstrate gate concentration, not automatically improved restoration.

G. Task 4: Face-to-sketch results

Sketch failure 1: photo1_image1600, style 1, PSNR 7.11 dB and SSIM 0.2531. Dense hair strokes and dark shading

TABLE X
Paired sketch reconstruction by style

Style	n	L1	PSNR	SSIM
1	619	0.0876	16.58	0.5006
2	381	0.1707	11.53	0.3654
3	46	0.0701	18.19	0.5725

are missing; the generated face is overly pale and smooth.

Sketch failure 2: photo1_image1599, style 1, PSNR 7.20 dB and SSIM 0.2569. Hair and facial contours are underdrawn compared with the dense reference sketch.

Sketch failure 3: photo1_image0837, style 2, PSNR 7.57 dB and SSIM 0.1889. The overall head shape is present, but textured hair/shading and fine feature lines are absent.

Sketch failure 4: photo1_image1034, style 2, PSNR 7.59 dB and SSIM 0.1537. The output face is pale and smooth, missing dark hair strokes and precise facial contours.

H. Observed failure modes and interpretation

Earlier dense and small spatial bottlenecks produced collapsed or blurry reconstructions. A larger 16-by-16 compressed representation with local residual refinement

soft: failures (test)

Clean target Input Output Absolute error
Egyptian_Mau_60 | occlusion medium | PSNR 11.63



german_shorthaired_96 | occlusion low | PSNR 11.88



staffordshire_bull_terrier_92 | occlusion high | PSNR 12.02



boxer_91 | occlusion high | PSNR 12.32



Fig. 16. Task 3: Soft mixture: four lowest-PSNR examples with distinct source images.

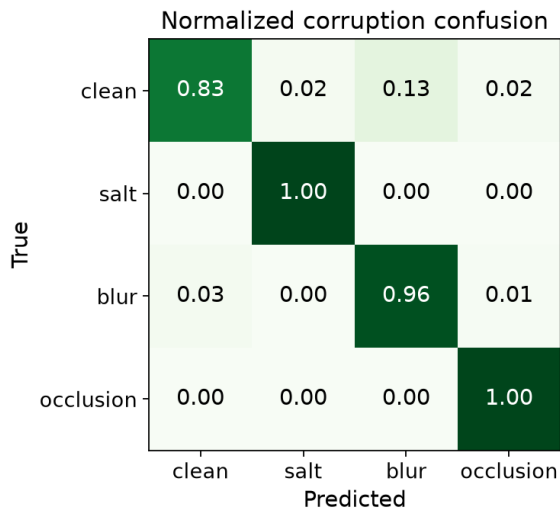
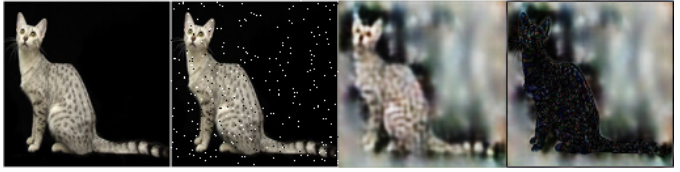


Fig. 17. Normalized confusion matrix; each row is a true class. Off-diagonal entries identify routing mistakes.

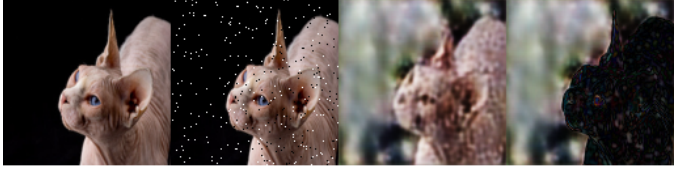
was investigated. A learning-rate 0.001 pilot saturated and was rejected. The subsequent lower-rate pilot reached

Classifier mistakes: hard routing

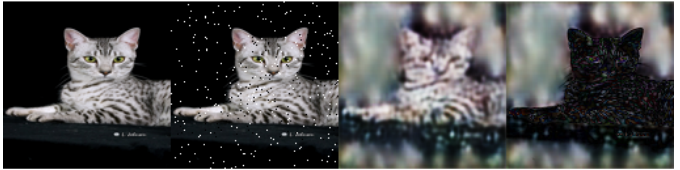
Clean target Input Output Absolute error
Egyptian_Mau_60 | true salt -> occlusion



Sphynx_23 | true salt -> occlusion



Egyptian_Mau_97 | true salt -> occlusion



Egyptian_Mau_63 | true salt -> occlusion

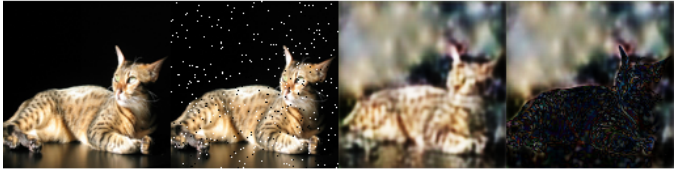


Fig. 18. Hard-routing examples with an incorrect classifier prediction. Targets and errors expose the restoration consequence.

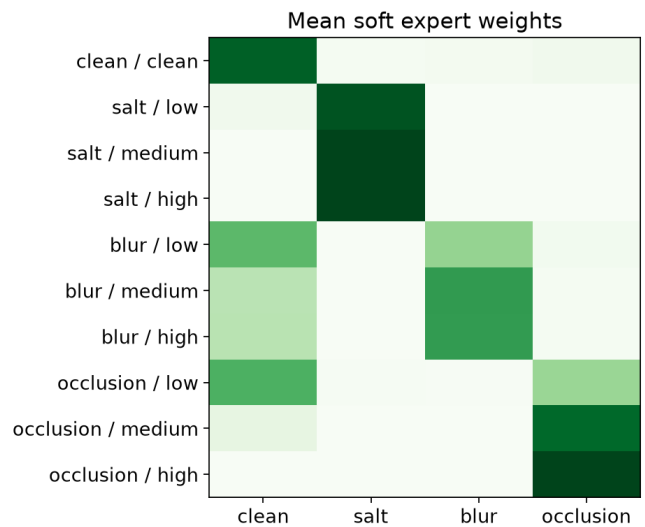


Fig. 19. Mean expert contribution by true condition and severity. Concentration in unrelated rows suggests imbalance.

validation objective 0.14389, compared with 0.24516 for

Sketch examples 1-4

Clean target Input Output Absolute error



photo1_image0788 | Style 2

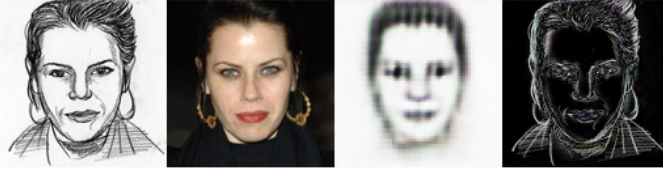


photo1_image0790 | Style 2



photo1_image0792 | Style 2



Sketch examples 5-8

Clean target Input Output Absolute error



photo1_image1397 | Style 1

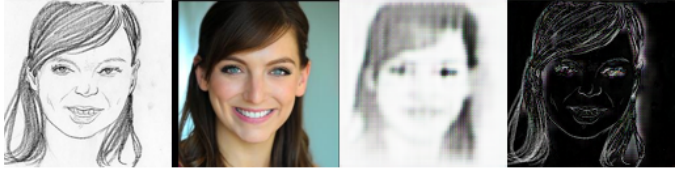


photo1_image1398 | Style 1



photo1_image1399 | Style 1

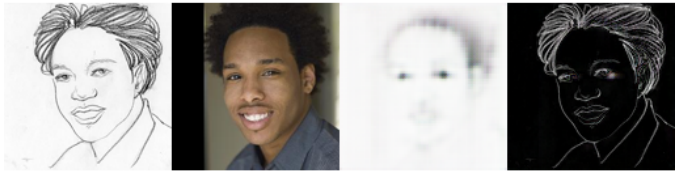


Fig. 20. Paired sketch examples 1-4: target, photograph, output and absolute error.

Fig. 21. Paired sketch examples 5-8: target, photograph, output and absolute error.

the previous selected model. This motivated a separate Optuna study and longer training, with all selection based on validation.

TABLE XI
Validation-led architecture pilots

Pilot	Epoch	Objective	PSNR	SSIM
Low LR	14	0.2074	16.56	0.4336
BatchNorm	5	0.2012	16.67	0.4465
Learned upsampling	10	0.1439	20.28	0.5786

These sequential pilots are exploratory, not controlled single-factor ablations: capacity, normalization, learning rate and upsampling differ. Their saved images and complete validation histories are retained. The selected learned-upsampling pilot used standard initialization; fresh final models use phase-matched ICNR initialization to address periodic artifacts, without guaranteeing their removal.

Both model sets use the same complete validation inputs and fixed objective $0.8L_1 + 0.2(1 - \text{SSIM})$. Numerical improvements must be considered with the per-condition

TABLE XII
Complete validation comparison, lower objective is better

System	Previous	Candidate	Change (%)
universal	0.2452	0.0853	-65.2
oracle_routing	0.2217	0.0704	-68.3
predicted_routing	0.2232	0.0710	-68.2
soft	0.1325	0.0642	-51.5
gan	0.1898	0.1853	-2.4

changes and saved images; a lower aggregate score does not establish improvement on every input.

The corruption classifier is retained from the preceding validation-selected run, while a normalized GAN with learned upsampling is tuned in a new study and trained from scratch. Restoration and routing quality are judged using condition-specific metrics and failure grids, not successful API execution alone.

Fresh CPU training from saved validation evidence after free Colab GPU quota; pilot weights were lost. The saved pilot measurements survived, but its weights did not. The improved models were trained afresh on the laptop CPU, while the retained classifier and exploratory pilots used the earlier free GPU runs. Local and Colab data manifests

Sketch examples 9-12

Clean target Input Output Absolute error

photo1_image1107 | Style 3



photo3_image0151 | Style 3



photo3_image0152 | Style 3



photo3_image0153 | Style 3



Fig. 22. Paired sketch examples 9-12: target, photograph, output and absolute error.

Face-to-sketch failures (test)

Clean target Input Output Absolute error

photo1_image1600 | Style 1 | PSNR 7.11

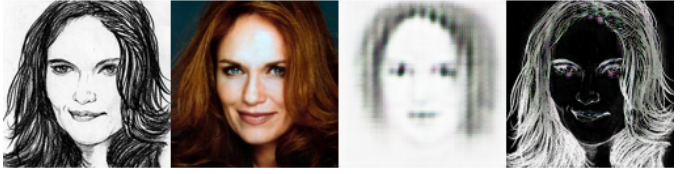


photo1_image1599 | Style 1 | PSNR 7.20

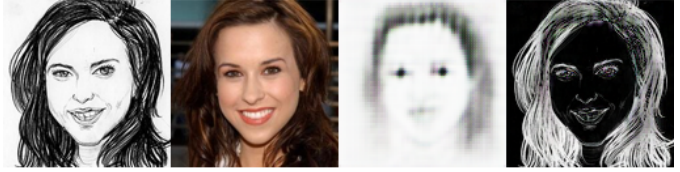


photo1_image0837 | Style 2 | PSNR 7.57

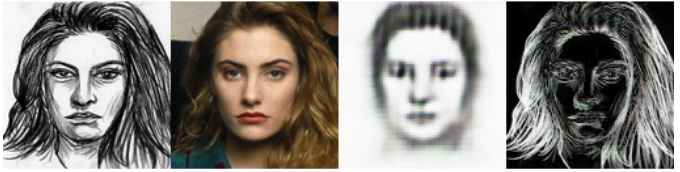


photo1_image1034 | Style 2 | PSNR 7.59



Fig. 24. Four lowest-PSNR paired sketch reconstructions, with distinct source photographs.

Photograph Style 1 Style 2 Style 3



Fig. 23. The same photograph under all three style conditions. Only its annotated style has a paired ground truth.

match after newline normalization.

The fixed final improvement schedule was universal=24 epochs, specialists=20 epochs, soft=12 epochs, gan=60 epochs. This deadline-aware budget was chosen before inspecting the final candidate test results.

I. Completed search and training records

universal: 2 completed trials, 1 pruned trials; best trial 0, validation objective 0.14329. The short-trial budget and complete search space are preserved in the study JSON.

classifier: 3 completed trials, 1 pruned trials; best trial 3, validation objective 0.32709. The short-trial budget and complete search space are preserved in the study JSON.

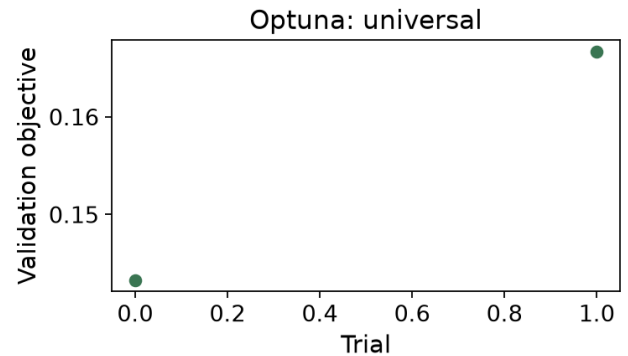


Fig. 25. Completed validation trial objectives for universal. These limited searches are not exhaustive.

specialists: 3 completed trials, 0 pruned trials; best trial 0, validation objective 0.13232. The short-trial budget and complete search space are preserved in the study JSON.

soft: 2 completed trials, 1 pruned trials; best trial 1, validation objective 0.06718. The short-trial budget and complete search space are preserved in the study JSON.

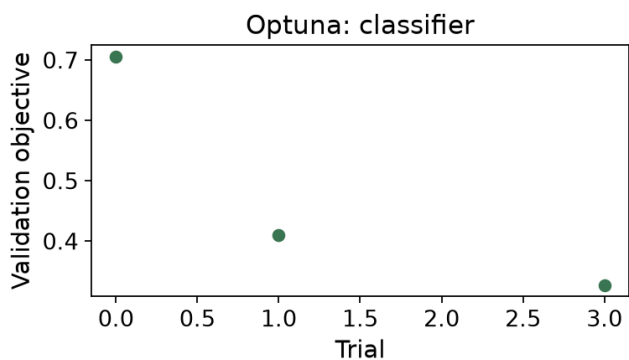


Fig. 26. Completed validation trial objectives for classifier. These limited searches are not exhaustive.

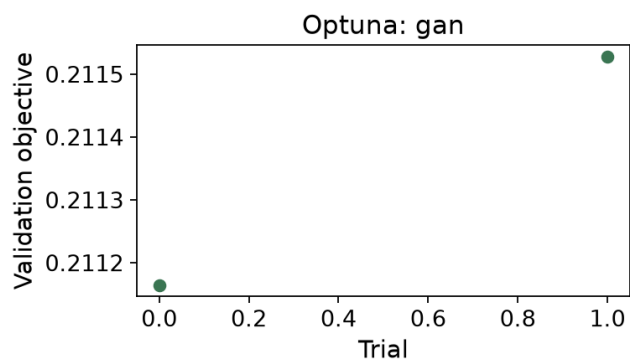


Fig. 29. Completed validation trial objectives for gan. These limited searches are not exhaustive.

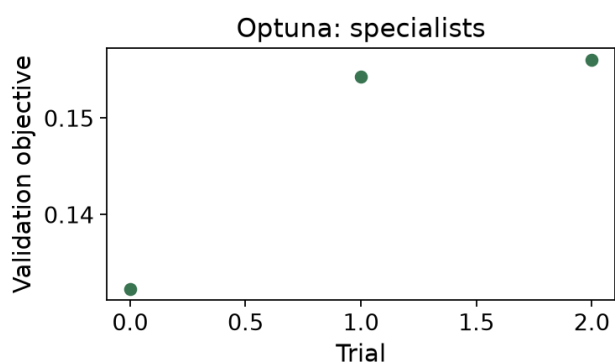


Fig. 27. Completed validation trial objectives for specialists. These limited searches are not exhaustive.

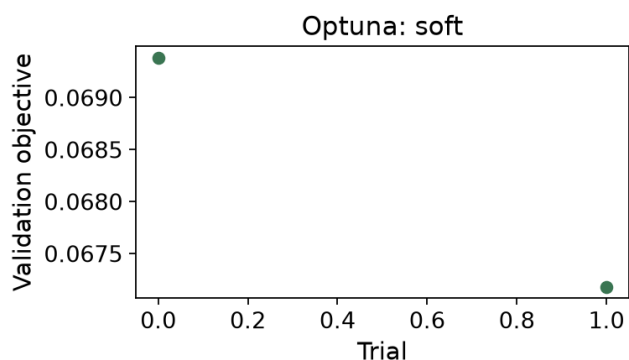


Fig. 28. Completed validation trial objectives for soft. These limited searches are not exhaustive.

gan: 2 completed trials, 1 pruned trials; best trial 0, validation objective 0.21116. The short-trial budget and complete search space are preserved in the study JSON.

Selected universal checkpoint: epoch 22. Configuration: base=24, latent=8192, dropout=0.0, batch_size=32, lr=0.0003, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1,

balance_weight=0.01, embedding=8, discriminator_lr=0.0002, l1_weight=100.0, detail=True, detail_bn=True, detail_shuffle=True.

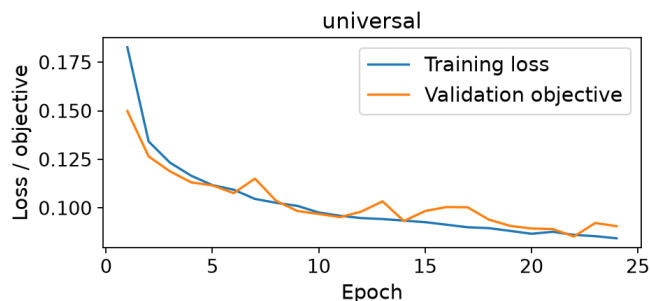


Fig. 30. Recorded training and validation objectives for universal; differing objectives may have different scales.

Selected classifier checkpoint: epoch 10. Configuration: base=24, latent=128, dropout=0.2896896099223678, batch_size=32, lr=0.0007896186801026692, weight_decay=0.0002661901888489054, alpha=0.8, temperature=1, classification_weight=0.1, balance_weight=0.01, embedding=8, discriminator_lr=0.0002, l1_weight=100.

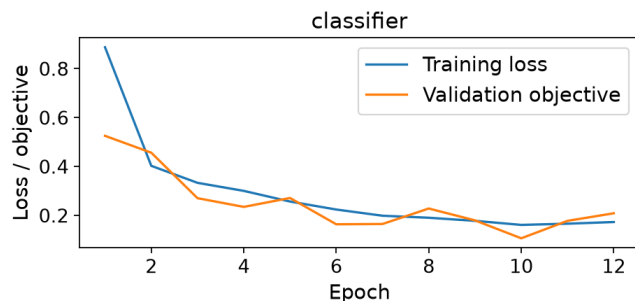


Fig. 31. Recorded training and validation objectives for classifier; differing objectives may have different scales.

Selected salt checkpoint: epoch 20. Configuration: base=24, latent=8192, dropout=0.0, batch_size=32, lr=0.0003, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1, balance_weight=0.01, embedding=8, discriminator_lr=0.0002, l1_weight=100.0, detail=True, detail_bn=True, detail_shuffle=True.

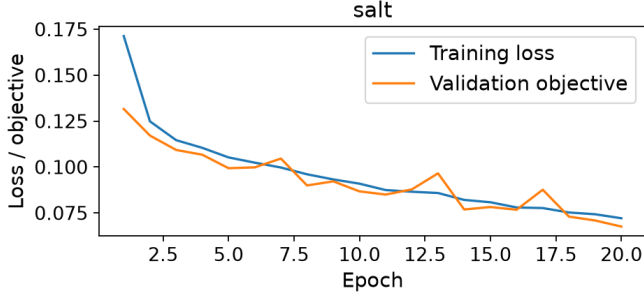


Fig. 32. Recorded training and validation objectives for salt; differing objectives may have different scales.

Selected blur checkpoint: epoch 18. Configuration: base=24, latent=8192, dropout=0.0, batch_size=32, lr=0.0003, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1, balance_weight=0.01, embedding=8, discriminator_lr=0.0002, l1_weight=100.0, detail=True, detail_bn=True, detail_shuffle=True.

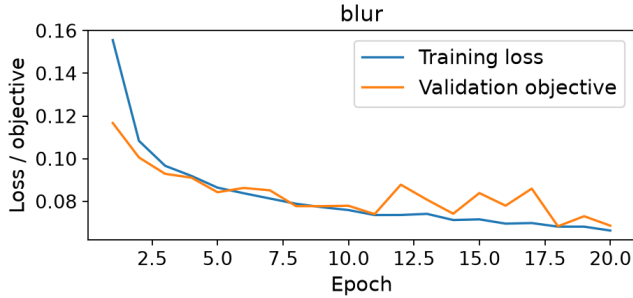


Fig. 33. Recorded training and validation objectives for blur; differing objectives may have different scales.

Selected occlusion checkpoint: epoch 18. Configuration: base=24, latent=8192, dropout=0.0, batch_size=32, lr=0.0003, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1, balance_weight=0.01, embedding=8, discriminator_lr=0.0002, l1_weight=100.0, detail=True, detail_bn=True, detail_shuffle=True.

Selected soft checkpoint: epoch 11. Configuration: base=16, latent=128, dropout=0.1, batch_size=32, lr=0.00027081608642499667, weight_decay=1e-05, alpha=0.6641915784487018, temperature=1.7486639612006325, classification_weight=0.035543498097189236, balance_weight=0.0023102018878452934, embedding=8, discriminator_lr=0.0002, l1_weight=100.0.

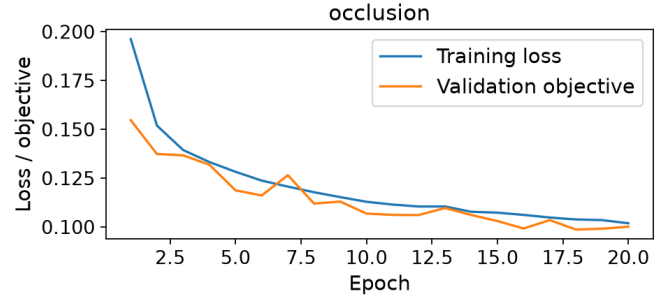


Fig. 34. Recorded training and validation objectives for occlusion; differing objectives may have different scales.

Selected gan checkpoint: epoch 12. Configuration: base=24, latent=128, dropout=0.05, batch_size=32, lr=0.0002, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1, balance_weight=0.01, embedding=16, discriminator_lr=0.0002, l1_weight=100.0, gan_refined=True.

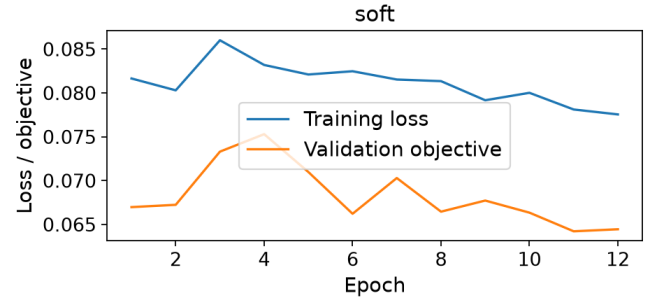


Fig. 35. Recorded training and validation objectives for soft; differing objectives may have different scales.

Selected gan checkpoint: epoch 12. Configuration: base=24, latent=128, dropout=0.05, batch_size=32, lr=0.0002, weight_decay=1e-05, alpha=0.8, temperature=1.0, classification_weight=0.1, balance_weight=0.01, embedding=16, discriminator_lr=0.0002, l1_weight=100.0, gan_refined=True.

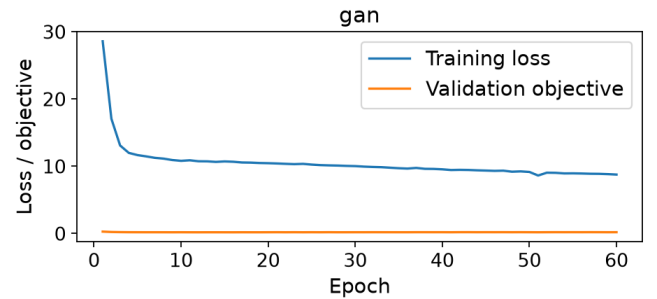


Fig. 36. Recorded training and validation objectives for gan; differing objectives may have different scales.

TABLE XIII
Mean test PSNR (dB) / SSIM per condition, averaged over severities (clean PSNR is capped at 100 dB)

System	Clean	Salt	Blur	Occlusion
Input (unrestored)	100.0/1.000	16.4/0.382	27.8/0.814	13.5/0.702
Universal	26.6/0.822	24.7/0.719	25.6/0.764	21.8/0.694
Oracle routing	100.0/1.000	26.0/0.796	26.0/0.791	21.8/0.708
Predicted routing	87.6/0.971	25.9/0.795	26.2/0.794	21.7/0.708
Soft mixture	50.7/0.995	26.1/0.810	27.4/0.828	20.4/0.749

X. Discussion of Results

Table XIII condenses the per-severity tables above. The paragraphs below interpret them; causal explanations are marked as hypotheses where no ablation was run.

a) Salt-and-pepper noise.: This is the clearest success. Relative to leaving the noisy input unchanged, mean PSNR rises by +8.3 dB (universal), +9.6 dB (oracle specialist), +9.6 dB (predicted routing) and +9.7 dB (soft mixture); specialist SSIM is 0.796 against 0.382 for the input. The dedicated specialist beats the universal model by 0.6 dB at low severity and 1.9 dB at high severity, consistent with the hypothesis that a single network sharing capacity across conditions does worse as noise grows. This is a comparison under our training budget, not a proof that specialisation is always superior.

b) Occlusion.: PSNR rises by +8.3 dB for oracle routing because black holes are filled with plausible colours. SSIM is mixed: at low severity the specialist SSIM (0.765) is below the unrestored input (0.857), because the autoencoder re-synthesises every pixel, including those the input already had exactly, whereas at high severity SSIM improves from 0.528 to 0.640. The failure grids show large masks across faces filled with smooth guessed colour, not recovered texture.

c) Blur.: This is the weakest condition. Mean PSNR change relative to the unrestored blurred input is -2.3 dB (universal), -1.8 dB (oracle), -1.7 dB (predicted) and -0.4 dB (soft). Oracle routing changes PSNR by -5.7, -0.1, +0.4 dB at low, medium and high severity, so restoration helps only at the highest severity and then by under one dB. A likely explanation is the reconstruction ceiling of the compressed latent: even on clean inputs the universal model reaches only 26.6 dB, while the blur specialist reaches 26.6 dB at low severity, where the unrestored input already scores 32.3 dB. Passing a lightly blurred image through the bottleneck is therefore likely to lower fidelity. No larger-latent ablation was run, so this is a hypothesis consistent with the data, not a demonstrated cause.

d) Clean inputs.: The universal model alters clean images (PSNR 26.6 dB, SSIM 0.822), a cost that hard routing and the soft mixture largely avoid through the identity branch: predicted routing keeps SSIM at 0.971, and the soft mixture at 0.995 because the gate gives the identity branch a mean weight of 91.1% on clean images.

e) Classifier and hard routing.: The classifier reaches 96.8% accuracy. Its errors concentrate in the clean class: clean recall is 0.829 and 12.5% of clean images are labelled blur. This is expected because low-severity blur (a 3×3 kernel with $\sigma = 0.7$) barely differs from a clean photograph; salt-and-pepper is almost always recognised (recall 0.997). Predicted routing differs from oracle routing by only -0.12 to +0.44 dB per condition on average, because few corrupted images are misrouted (for example 3.1% of blurred images are labelled clean, where bypassing the specialist can even help low-severity blur). Misrouting can nevertheless be severe: the four worst predicted-routing cases are salt-and-pepper images sent to the occlusion expert (PSNR 7.1–8.8 dB against 20.0–28.6 dB with the correct expert); the saved failure analysis notes dark backgrounds.

f) Soft gate.: For salt-and-pepper the gate is nearly one-hot (salt expert weight 0.953 at low severity and at least 0.9996 at medium and high severity). For blur and occlusion at low severity it shares weight with the identity branch (identity weights 0.56 and 0.59). This explains why the soft model beats the specialists on low-severity blur (29.9 versus 26.6 dB) yet fills less of the mask at low-severity occlusion (20.3 versus 23.6 dB PSNR, 0.860 versus 0.765 SSIM). No branch contributes under 1% of the global mean weight, so no expert is inactive, and no unrelated expert exceeds a mean weight of 0.043, so none dominates unrelated inputs. The price is compute: the soft graph evaluates all four branches, with single-request CPU inference of 25.4 ms against 12.4 ms for hard routing and 12.9 ms for the universal model.

g) Face-to-sketch.: Mean SSIM over the 1,046 test pairs is 0.455; style 2 ($n = 381$) is weakest at 11.5 dB PSNR, while style 3 has only 46 test pairs, so its score is noisy. The generated sketches reproduce head pose and outline but are pale and smooth, without the hatching of the references (see the failure analysis). Discriminator real and fake losses stay between 0.36 and 0.83, so neither network overwhelms the other. The validation objective is lowest at epoch 12 (0.185) and is 0.201 at epoch 60: longer adversarial training did not improve the L1/SSIM-based objective, so the epoch-12 checkpoint is deployed. Plausible causes are the small generator, only 899 training pairs at 128×128 , and the weight-100 L1 term favouring averaged outputs; none was isolated by an experiment.

h) How the evidence shaped the design.: Validation pilots decided the architecture. The validation objective of the first dense-latent model was 0.322; an 8×8 spatial latent reached 0.281, a 16×16 latent with local residual blocks 0.207, and the final design (8,192-value latent, BatchNorm, PixelShuffle upsampling) 0.144 in its pilot and 0.085 after full training. These runs differ in budget, so the sequence indicates direction, not a controlled ablation. The remaining weaknesses above, blur and sketch detail, are the points where additional capacity, a sharper loss or a longer GPU schedule would be tried next.

XI. Application Architecture and Deployment

The React/Tailwind interface provides four workspaces in one application. FastAPI validates JPEG, PNG and WebP uploads, rejects files over 10 MB and oversized decoded images, applies optional reproducible corruption, and runs ONNX Runtime CPU inference. It returns input/output PNGs, route probabilities or soft weights, and inference latency. Runtime corruption makes the clean reference available for an error map; an arbitrary uploaded corrupted image does not provide an assumed ground truth.

TABLE XIV
Backend endpoints

Endpoint	Purpose
GET /api/health	Per-model and per-workspace readiness
GET /api/samples	Clean sample images
GET /api/experiments	Test metrics and MLflow run records
POST /api/corrupt	Reproducible corruption preview
POST /api/universal-restoration	Task 1 inference
POST /api/hard-routing	Task 2: probabilities, expert, output
POST /api/soft-mixture	Task 3: four weights, output
POST /api/face-to-sketch	Task 4: photo and style to sketch

The frontend is served by Nginx and proxies API requests to the backend. Docker Compose starts both containers using docker compose up --build; the browser URL is <http://localhost:8080>. Model and result directories are mounted read-only. Session initialization is excluded from inference timing. Missing models, synthetic test checkpoints and failed checksum verification produce a clear unavailable-model state rather than substitute output.

ONNX export checks random, black and white inputs for batches of one and two, including all three GAN styles. Acceptance uses absolute tolerance 10^{-4} and relative tolerance 10^{-3} for every output. Actual maximum deviations are stored in the deployment manifest. The complete soft inference pipeline is one graph; the classifier and specialists are separate graphs for hard routing.

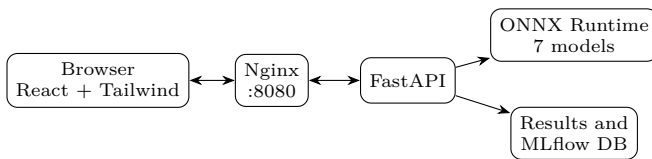


Fig. 37. Deployment: two Docker Compose services; model and result files are mounted read-only into the backend.

Local verification passed all seven model checksums and finite-output checks, all four API workspaces, and all three sketch styles using CPUExecutionProvider. Measured sample-request inference times ranged from 10.66 to 25.43 ms. These are single-request checks, not a throughput benchmark. The universal black-versus-white output

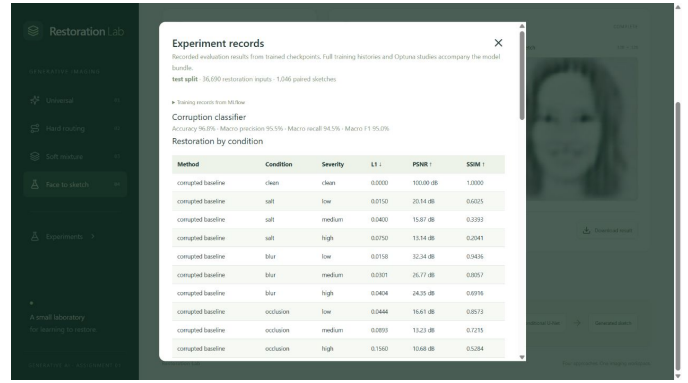


Fig. 38. Application with imported trained ONNX models: app-experiments

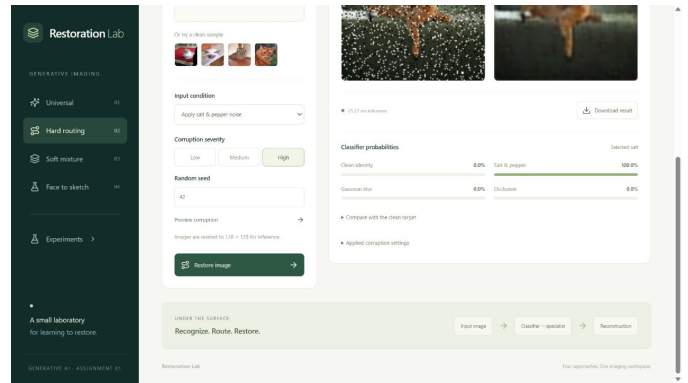


Fig. 39. Application with imported trained ONNX models: app-hard

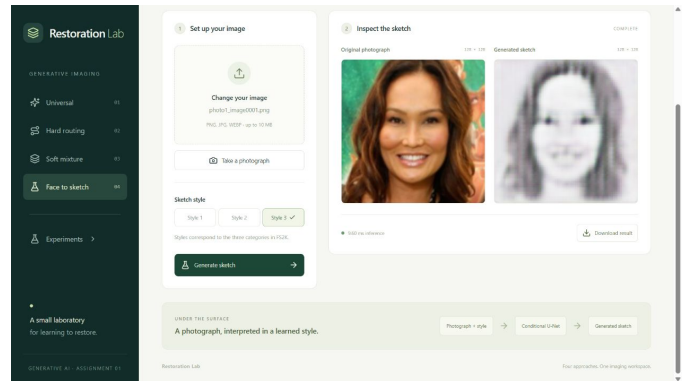


Fig. 40. Application with imported trained ONNX models: app-sketch

difference was 9.79e-01. This confirms input sensitivity for these two controls, not restoration quality.

Stitch's HTML and PNG exports are retained in the repository (docs/stitch-evidence). The implemented interface was reconciled with this design: dark navigation rail with the four workspaces, input panel, result panel, error and experiment views, and a green palette. Mock-up metrics and class names that correspond to no implemented function (for example the A100 latency badge and the ViT parameter counts) were omitted from the working

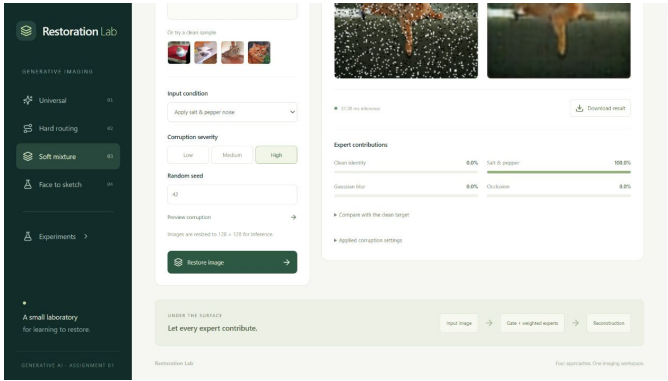


Fig. 41. Application with imported trained ONNX models: app-soft

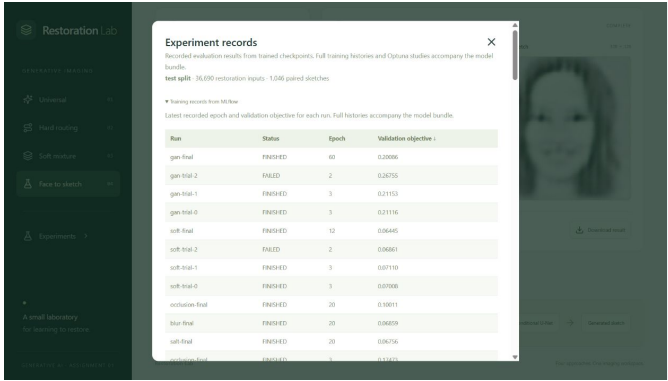


Fig. 42. Application with imported trained ONNX models: app-tracking

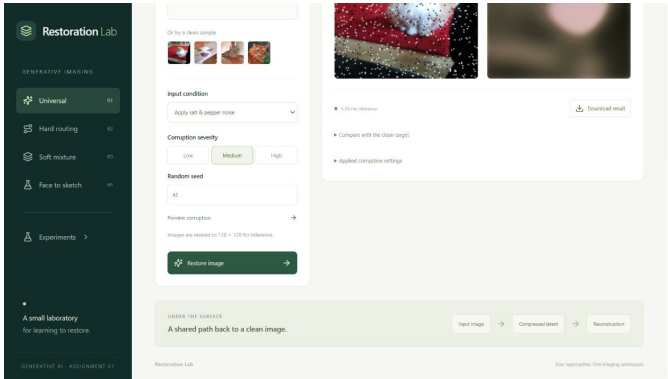


Fig. 43. Application with imported trained ONNX models: app-universal

application.

Status: Docker Compose was built and executed with a Linux Docker engine. HTTP tests passed for all four workspaces and all three sketch styles at localhost:8080.

A. Links

Repository: <https://github.com/Abdullah73/genai-assignment-1>.

Original Stitch design: <https://stitch.withgoogle.com/projects/16111510614845497849>.

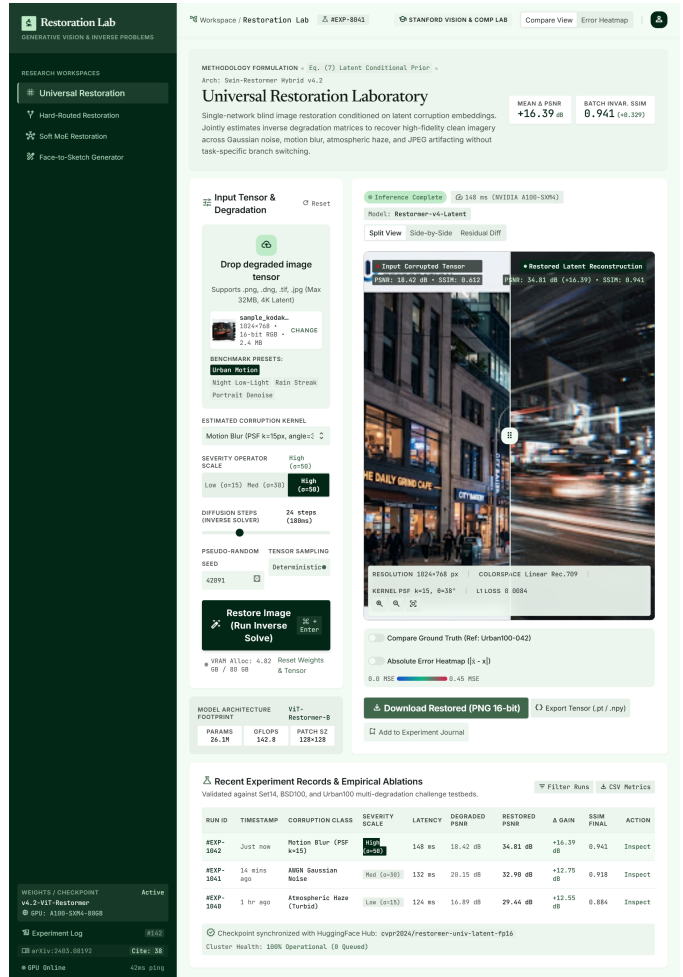


Fig. 44. Original Google Stitch universal-workspace design. Decorative metrics and classes in this mockup are not experience results.

XII. Limitations and Conclusion

The main limitations are the following. (1) The compact bottleneck discards fine texture, which appears to cap reconstruction near the level reached on clean inputs, so blur restoration does not exceed the uncorrected input on average. (2) Optuna searches used short trials (three epochs, 40 batches) and few trials, so the selected configurations are the best of a small search, not a global optimum; in three studies the best trial was the seeded starting configuration. (3) The classifier confuses clean and low-severity blurred inputs, and the soft gate shares that ambiguity. (4) The sketch generator produces smooth, pale sketches without hatching; style 3 has only 46 test pairs, so its scores are noisy. (5) Single-corruption training does not establish reliable mixed-corruption restoration, and disjoint-band masks do not cover all real occlusion geometries. (6) Inference is 128×128 on CPU; larger uploads are resized. (7) Final training used a CPU after free GPU quota was exhausted, which limited schedule length. (8) The soft mixture keeps its warm-up learning

rate after unfreezing the experts instead of lowering it, as the assignment suggests; the effect of such a drop was not tested. Future work should enlarge the latent for blur, add a sharpening or perceptual loss, run longer GAN schedules with more Optuna trials on a GPU, and evaluate mixed corruptions. The measured results above characterize the selected compact models under the recorded training budget; broader claims require further experiments.

Appendix A AI Use and Verification

TABLE XV
AI tools used, purpose and verification

Tool	Used for	How outputs were checked
OpenAI Codex	Reading the assignment, finding sources, drafting models, training and evaluation code, FastAPI/React code, Colab notebook, tests, first report draft	20 automated tests (corruptions, balanced batches, paired augmentation, SSIM, bottleneck, gradients through all mixture branches, style conditioning, upload validation, resume, ONNX), validation grids, ONNX-vs-PyTorch comparison, Docker runs
Anthropic Claude (Claude Code)	Final audit of the repository against the assignment, report revision (protocol, discussion, limitations), documentation and repository packaging	Numbers in the discussion are computed from the saved result files by the report script; documents compared with files; tests re-run
Google Stitch	Original interface design (exports retained in the repository)	Controls compared with implemented functions; decorative mock-up metrics omitted

A soft-mixture dropout-mode mismatch found by ONNX consistency testing was corrected. Synthetic-data tests verify mechanics only; dataset performance comes from saved real checkpoints evaluated on deterministic manifests. The student remains responsible for understanding and explaining every component.

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